



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/AGSR Division

FIRST SEMESTER 2026-2027

Course Handout (Part II)

Date: 2 Aug 2026

In addition to Part-I (General handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

COURSE NO. : CS F342
COURSE TITLE : **COMPUTER ARCHITECTURE**
INSTRUCTOR-IN-CHARGE : **VIRENDRA SINGH SHEKHAWAT**
E-mail: vsshekhawat
Instructor(s) : Tejasvi Alladi
E-mail: tejasvi.alladi
Lab Instructors : Ashutosh Bhatia (E-mail: ashutosh.bhatia)
: Nivedan yakoli (E-mail: p20230032)
: Aniket Upadhyay (E-mail: p20241007)
: Vijay Shankar A (E-mail: p20240905)
Course page: <https://nalanda.bits-pilani.ac.in>

Scope and Objectives:

This course develops a comprehensive understanding of computer architecture and organization, from quantifying processor performance to designing a working processor and its supporting memory system. It begins with performance measurement and comparison using key metrics, benchmarking tools, and Amdahl's law. Building on this, it covers the MIPS architecture and instruction set. The course then examines the design of single-cycle and multi-cycle processor. It further explores pipelined processor design, the handling of structural, data, and control hazards through forwarding, stalling, reordering, and branch prediction techniques. The memory hierarchy is studied to exploit locality of reference, covering memory technologies, cache mapping functions, cache controller design, cache performance metrics, and multi-level caches. Finally, the course introduces advanced techniques for increasing instruction-level parallelism, including multiple issue, loop unrolling, speculation, and dynamic scheduling.

TEXT BOOKS

(T1) Patterson, David A & J L Hennenssy, *Computer Organisation & Design*, Elsevier, 6th Ed., 2021.
(T2) Samir Palnitkar, *Verilog HDL: A Guide to Digital Design and Synthesis*, Pearson Education Asia, 2nd Ed. 2006.

REFERENCE BOOK

(R1) J.L. Hennessy & D.A. Patterson, *Computer Architecture: A Quantitative Approach*, Morgan Kauffmann, 6th Ed, 2019.



Please Do Not Print Unless Necessary





BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/AGSR Division

Module No.	Module Name	Learning Objectives
M1	Computer System Performance and Its Measurement	In this module, you will learn - the performance measurement of a computer. - the performance comparison of two computers by understanding key metrics and utilizing various benchmarking tools. - the computer performance enhancement using Amdahl's law.
M2	MIPS Architecture and Instruction Set	In this module, you will learn - the key elements of the Microprocessor without Interlocked Pipelined Stages (MIPS) architecture assembly instructions. - the MIPS architecture's ISA, which includes instruction formats (R, I, and J type), memory structure, addressing, and a register file. - the step-by-step process of designing an ALU by adding the capability of executing arithmetic, logical, memory reference, and branch instructions.
M3	Data-path and Control Design: Single Cycle and Multi-cycle processor Architectures	In this module, you will learn - to combine the different data-path components of an MIPS processor into a single data-path and add the control to complete the single-cycle processor implementation? - about different types of control signals required to be enabled from the control unit to execute instructions by the data-path components. - how to break up instructions into multiple steps and execute them in multiple cycles? - how to design a multi-cycle data-path and its control unit for instruction execution?
M4	MIPS Pipeline Architecture Design and Branch Prediction Techniques	In this module, you will learn - how to overlap the execution of multiple instructions in a pipelined processor to enhance its performance? - about the different situations in which simultaneous execution of instructions is leads to inconsistent results. - how to design the data-path and control for a pipelined processor for the MIPS instruction set? - how to handle data and control hazards in pipelining data-paths to avoid inconsistency in the results and reduce stalls in the pipeline.





BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/AGSR Division

		• about data forwarding, stalling, and reordering of instructions for handling hazards in the pipeline. • about static and dynamic branch prediction techniques to improve pipeline speedup by reducing stalls in the pipeline data path.
M5	Memory Hierarchy in Computing Systems	In this module, you will learn • how to organize a computer system memory at different levels to exploit the locality of reference for improving the performance? • about different types of memory and their characteristics. • about the different mechanisms to map main memory locations to cache memory for consistency? • how to measure and improve the performance of cache memory by making changes in block size, number of levels of cache memory, and associativity?
M6	Advanced Concepts in Pipelining	In this module, you will learn • about the techniques like multiple issue, loop unrolling, speculation which are used for further increasing instruction level parallelism to get more speedup.





PLAN OF STUDY

Lecture No.	Topics	Reference to T1
M1: Computer System Performance and Its Measurement		
1-3	Computer performance and Performance equations, MIPS rate, Amdahl's law, Standard Performance Evaluation Corporation (SPEC) Benchmarks	Ch. 1.3-1.6, 1.9, 1.11
M2: MIPS Architecture and Instruction Set		
4-10	MIPS Instructions: Arithmetic, Logical, and Branch Instructions, Floating-Point representations and FP Arithmetic, FP computation capability in MIPS, ALU design, Register-file design.	Ch.2, Ch. 3.1-3.5
M3: Data-path and Control Design: Single Cycle and Multi-cycle processor Architectures		
11-20	Single-cycle data-path design, Single-cycle control design, multi-cycle data-path design, multi-cycle control design, Hardware vs. Microprogrammed control	Ch.4.1 – 4.5, Appendix - D
M4: MIPS Pipeline Architecture Design and Branch Prediction Techniques		
21-30	Pipeline data-path design, Pipeline Control Signals and Pipeline Control Implementation, Pipeline Hazards: Structural, Control and Data, Hazard handling Techniques: Data forwarding, Stalling, Delayed branch, Exception handling in pipeline, Dynamic branch prediction techniques: Correlating and tournament predictors.	Ch. 4.6 – 4.10 + Class Notes
M5: Memory Hierarchy in Computing Systems		
31-36	Memory technologies, Exploiting Memory Hierarchy in Computing Systems, Cache Memory and Locality of References, Cache mapping functions: Direct, Associative, Set-Associative, Cache controller design, Cache Performance Metrics, Multi-level caches, Victim cache.	Ch. 5.1 – 5.5, 5.12
M6: Advanced Concepts in Pipelining		
38-40	Hardware and Software techniques to increase Instruction Level Parallelism (ILP). Loop Unrolling and Dynamic Scheduling using Tomasulo algorithm.	Ch. 4.11





BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/AGSR Division

EVALUATION SCHEME:

S. No.	Component	Duration	Weightage	Date and Time	Nature of component
1.	Mid Semester Exam	1.5 hrs	30%	7-10-2026 (AN1)	Close Book
2.	Lab Participation	---	03%	NA	Open Book
3.	Lab Exam	2.0 hrs	22%	To be announced	Open Book
4.	Comprehensive Exam	3.0 hrs	45%	10-12-2026 (FN)	Close Book

Notices: All notices will be posted on the **NALANDA LMS**.

Makeup Policy: In genuine cases, on a case-by-case basis, makeup shall be allowed. Prior permission from the instructor-In-charge is a must. There will be 2 buffer slots for Lab Participation component.

Lab Sessions: Lab sessions will focus on designing a MIPS processor component using Verilog-HDL programming.

Chamber (6111-C @IT Services Office) Consultation Hour: Thu 5:00 – 6:00 PM

Instructor-in-charge
CS F342



Please Do Not Print Unless Necessary

